

Automatic Measurement of Fetal Head Circumference from Ultrasound Images using Deep Ensemble Transfer Learning Regression Model

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Abstract—A key biometric for tracking fetal development throughout pregnancy is fetal head circumference (HC). This measurement is usually obtained through ultrasound images by manually adjusting the fetal ellipse. As a manual experiment is operator-reliant, errors in intra- as well as inter-variability are likely to occur. Therefore, fetal head segmentation has been attempted using automated segmentation methods, particularly those based on recent deep encoding-decoding systems. Conversely, we recommend a regression deep ensemble convolutional neural network model to automatically calculate the head circumference, eliminating the requirement for manually annotated segmented images or handcrafted features. The regression models were used for training and testing upon the HC18 dataset, including images from all pregnancy stages. We investigated the performance of various pretrained CNN architectures, considering regression-appropriate loss matrices to choose the three best models, and finally, developed an ensemble regression model. The obtained results show a significant improvement in HC prediction by the proposed ensemble regression model alongside a mean absolute error of 1.05 ± 1.22 , which is found superior to its base models as well as the conventional segmentation method.

Keywords—fetal head circumference, CNN deep ensemble regression model, CNN-based segmentation, ultrasound images

I. INTRODUCTION

Automatically assessing the circumference of the fetal head is an essential biometric for monitoring fetal development and estimating gestational age during pregnancy. Traditional measurements are taken in a clinical context by sonographers using ultrasound (US) images to physically trace the form of the skull and align it with an ellipse. However, manual contouring is a process that depends on the operator and is susceptible to both intra- and inter-variability, leading to inaccurate data. The intra-operator variability has been determined to have 95% limits of agreement at ± 7 mm, whereas the inter-operator variability is ± 12 mm as reported in literature [1]. It is challenging to recognize the head contour in US images owing to the insufficient signal-to-noise ratio for the speckle noise, and with that, the often blurred or missing borders of the contours.

Nowadays, automated segmentation methods are commonly employed to assess the fetal HC in ultrasonography images using the proven power of different image processing techniques [2]-[4]. Segmentation methods often incorporate various post-processing approaches to adjust the outcomes to fit an elliptical shape. This procedure has numerous steps, is ad hoc, and can be error-prone. However, segmentation step

can be useful to calculate feature from the image for further uses towards classification task with sufficient computational costs. For instance, volume measurement and staging of lung tumor from the segmented CT slices using conventional image processing supported Fuzzy C-means clustering method [2]. Volume estimation of dental caries was carried out from K-means clustering and threshold-based for segmented regions merging of dental computed tomography slices [3]. In our previous study, we have segmented the fetal brain plane using UNet to classify the trans-cerebellum, trans-thalamic, and trans-ventricular planes efficiently using deep transfer learning [4]. In contrast to conventional segmentation-based approaches, segmentation-free approaches avoid the complications and can use deep learning to discover direct connections between image characteristics and the HC. Instead of segmenting, which is an expensive and susceptible to errors process, one might try to evaluate the traits or biomarkers directly. The advancement of intelligent machine learning, which empowers end-to-end relapse and leverages the quality of computer-aided measurement, has given this field a moment term on life [5]-[9].

We intend to investigate the possible advantages of estimating the HC directly from ultrasound images without segmentation and compare it to the conventional segmentation approach. Direct, segmentation-free algorithms use less information to predict biomarkers and may not achieve the same accuracy as segmentation-based approaches using just a CNN regressor, though some works on single transfer learning-based CNN regression model [7], [10] reported for different applications. Being motivated by the superior performance of ensemble learning, in this study, we directly estimate the fetal HC from US images using an ensemble deep CNN-based regression model without resorting to segmentation to investigate the performance of ensemble CNN regressor which might not be used earlier for such HC estimation objective from US images according to our understanding. The major contributions in this work are:

- A deep ensemble CNN regression model based one-stage approach is proposed and investigated to estimate the fetal HC from ultrasound images.
- In this one-stage approach, there is no need to segment the head contour and perform ellipse fitting to measure the HC.
- An ensemble regression-based CNN model, including well-known EfficientNetB2, ResNet-50V2, and

MobileNetV2 models with averaging is designed to estimate the HC.

- A rigorous comparative study is also conducted, considering related techniques using a publicly accessible data set of HC18, and found superior performance for the desired model.

The remainder part of the current study is arranged as follows: a summary of recent research on the subject is given in the related works segment. The methodology section describes the dataset, approach, and model assessment procedure. The results analysis is part of the outcome analysis. The entire research investigation is finished in the conclusion section.

II. RELATED WORKS

In earlier research, skull edge characteristics have been extracted using various machine learning-based techniques, including Haar-like features and various classifiers [11]- [14]. Creating a classifier based on random forests using Haar-like features is the first stage [11] in using machine learning to locate the fetal head. After that, an elliptical fitting and the transform of Hough are used to quantify the HC. Texton-based fetal head segmentation using filter bank [15], multilevel thresholding approach [16], and contour fragment model [17] are further techniques for obtaining characteristics of the border or head zone. After the skull characteristics were recovered, several techniques, including the Hough transform and elliptic fitting, were used to fit the contour of the skull boundary and estimate HC [18]. While some positive outcomes have been obtained using the methods mentioned above, each of them necessitates prior information or artificially created features that are computationally intensive and have low robustness. Convolution-based artificial neural platforms have achieved appeal in the disciplines of form evaluation, segmentation, and medical picture analysis in current years. Z. Sobhaninia et al. measured biometric parameters in fetal US using multi-task deep learning [19]. Ref. [20] proposes a Mask-R2CNN artificial neural network for performing HC size field projection for head distinguishing thoroughly without the need for previous HC localizations or additional processing for outlier elimination. The authors of [21] segment the HC using a regression-based CNN based on proximity fields after initially using an area suggestion CNN for head localizations. However, these methods have large networks and are laborious because several complex post-processing steps, including expansion, edge extraction, and corrosion, are performed to obtain the skull boundary region pixels after predicting the skull region. Lastly, HC is measured using ellipse parameter fitting. The segmentation outcome significantly affects the head circumference measurement accuracy, and it is also challenging for ultrasonic images with ambiguous or partial boundaries. Different custom and pretrained CNN regression models, such as Reg-CNN_1M and Reg-CNN_263K, Res-VGG16, and Reg-ResNet50 models have been used in [7].

The methods mentioned above have majorly focus on head localization, segmentation, and ellipse adaptation to assess the head circumference, or automatic estimation of HC from ultrasound images using different single CNN regression models. Though the regression based direct estimation showed promises considering the benefits of using less information to predict the biomarkers, improvement of direct segmentation-free methods might be done through ensemble

of suitable regression models. In this study, we proposed a deep ensemble regression model, which is a direct approach to measure the HC avoiding the segmentation stage.

III. METHODOLOGY

This study proposes a method for determining the fetal head circumference from fetal ultrasonography. By combining three well-known CNN architectures, EfficientNetB2, ResNet-50V2, and MobileNetV2, that were selected from a range of pre-trained models' performances, a deep ensemble regression model was created to measure the fetal HC. The general framework of the suggested ensemble regression method of estimating the HC is shown in Fig. 1. The following subsection covers data preprocessing, segmentation-free deep ensemble regression of CNNs, segmentation-based approach, and regression loss calculation.

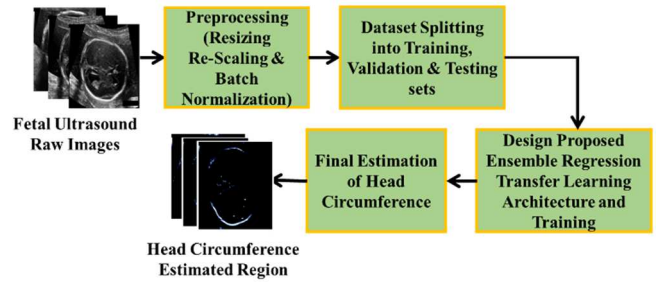


Fig. 1. Proposed framework of fetal HC measurement using ensemble CNN regression model.

A. Data Description and Preprocessing

Two-dimensional shapes (2D) ultrasonography images of prenatal head circumference is obtained from the Department of Obstetrics database at Radboud University Medical Centre in Nijmegen, the Netherlands [1], [22]. The dataset included 1334 two-dimensional images taken by 551 expectant mothers who underwent routine ultrasound examinations from May of 2014 to May of 2015. A skilled sonographer annotates the fetal head's baseline outline as an ellipse shape, and each image's pixel size and HC value are contained in a text file. This dataset covers 10 to 40 weeks of gestation [22]. The preprocessing step involves normalization and resizing the images from 800×540 pixels to 224×224 pixels. To improve convergence and to normalize the HC values, divide them by the largest HC value. We organized the collected data into three parts: the training portion, the validation portion, and the blind test portion in an 80:10:10 ratio.

B. Head Circumference Measurement based on Deep Ensemble Regression

1) *CNN Regressor*: Conventional CNNs consist of many convolutional layers, an identification SoftMax layer and fully connected layers at the end. When converting a CNN design for classification to regression, in place of the SoftMax layer, a fully linked regression layer with sigmoid or linear activation is used. A linear transfer function produces an activation that is proportionate to the input and not limited to a certain range. The following is an expression for a multiple CNN linear regression equation:

$$Y = a + b_1X_1 + b_2X_2 + e \quad (1)$$

where $b_1, b_2, \dots$ are the slopes of $X_1, X_2, \dots$ are the independent variables, and e is the error term.

2) Proposed Ensemble Deep Transfer Regression Model:

In the last decades, deep learning regression has significantly extracted biomarkers directly from medical images. The intermediate state, known as segmentation, and additional ad-hoc post-processing processes that may be computationally costly for both model training and image annotation can be readily eliminated by employing the regression-based CNN model [8]. Recently, regression CNNs are crucial to various computer vision fields, including head-position estimation, human-body pose estimation, and face landmark identification etc.

For object detection utilizing images as input data, popular pretrained deep CNNs like VGG16, Inception-V3, ResNet-50, Xception-V3, MobileNetV2, DenseNet201, EfficientNetB2, and numerous others are using enormously. We evaluated the performance extensively of the selected EfficientNetB2 [23], ResNet-50V2 [24], and MobileNetV2 [25] individually and after performing the ensemble as shown in Fig. 2. We have trained various models on the dataset and assessed their performance. Lastly, for our objective, we have selected the top three models: EfficientNetB2, ResNet-50V2, and MobileNetV2. On three models, we substituted a new layer for the ultimate fully connected layer. A global average pooling layer, seven dense layers of 1024, 512, 256, 128, 56, 32, and 1 neuron in ascending order, and an activation function known as ReLU are its constituent parts. For estimating head circumference, it additionally has an output layer using a linear activation coefficient and two dropping layers with a possibility of 0.5. ResNet-50V2 [25]. The architecture of EfficientNetB2 was based on the Efficient module, which contains 3×3 filter sizes pooling layers in addition to three kernel adjustments (1×1, 3×3 & 5×5) sizes of convolution layers. MobileNetV2 has been deployed along with ResNet-50V2 and EfficientNetB2 as well. The ensemble learning architecture with these CNNs with averaging is displayed in Fig. 3, and the rule of ensemble averaging is illustrated in (2).

$$\sum_{t=1}^T d_{t,j} = \frac{C}{T} \sum_{t=1}^T d_{t,j} \quad (2)$$

$d_{t,j} \in \{0,1\}$, $t = 1 \dots T$ and $j = 1 \dots C$; wherein T is the entire variety of learners and C represents the class numbers.

3) *Regression Loss Metrics*: The Huber loss (HL), mean absolute error (MAE) loss, and mean square error (MSE) loss are among the loss functions frequently used to assess the regression CNN which are mathematically defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (p_i - g_i)^2 \quad (3)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |p_i - g_i| \quad (4)$$

$$HL = \begin{cases} \frac{1}{n} \sum_{i=1}^n \frac{1}{2} (p_i - g_i)^2, & \text{for } |p_i - g_i| < \delta \\ \frac{1}{n} \sum_{i=1}^n \delta * (|p_i - g_i| - \frac{\delta}{2}), & \text{otherwise} \end{cases} \quad (5)$$

Here, p_i refers to the probabilities of expected pixels, the pixel number of an image is n , delta is a hyperparameter, g_i is the brain circumference's true worth in pixels, and is empirically set to 1.

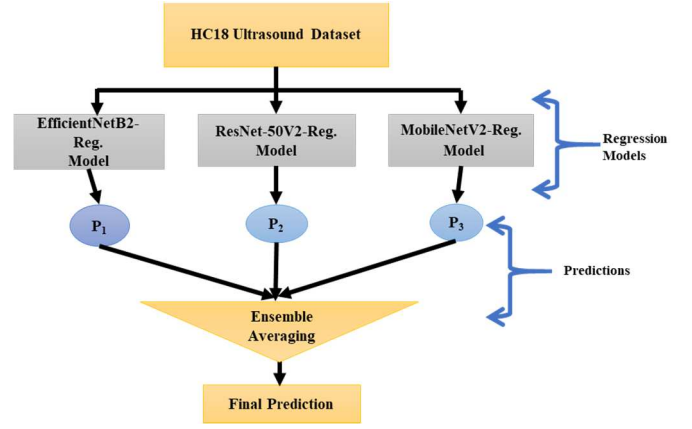


Fig. 2. Proposed ensemble learning model based on EfficientNetB2, ResNet-50V2, and MobileNetV2 models.

C. Head Circumference Measurement based on the Conventional Segmentation Approach

We have intended to compare the regression-based direct HC estimation with the segmentation-based indirect conventional approach. Different deep learning-based segmentation techniques are now dominantly used for medical image segmentation. U-Net [26], LinkNet [27], PSPNet [28], and SEGNet [29] are a few well-known models. A typical segmentation-based approach is shown in Fig. 3.

We have chosen the SEGNet [29] concept to predict the HC to compare the results of the segmentation-based concept with our proposed ensemble deep CNN regression-based concept. We have named the segmentation network as SEGNet, considering its regression-based counter model, which will be named with the Reg_ prefix. The SEGNet adds a pixel-wise grouping layer that comes after the encoder and with that decoder networks. Through the encoder platforms, the first 13 layers of the VGG16 network, used for identifying objects, are replicated by 13 convolutional layers [30]. A decoder network of 13 layers is formed by matching each encoder depth with a commensurate depth in the decoder. Max-pooling is carried out using an element-wise adjusted uniform non-linear (ReLU) max (0, x) and a 2×2 frame with a width of 2. Next, a factor of two is applied to down-sample this output. An accessible decoding filter branch processes the sparse directions, producing dense map representations that are subsequently batch-normalized. While the initial encoder accepts an input with three channels (RGB), the decoder produces a feature map with multiple channels. Conversely, the feature maps from other decoders in the network are consistent with their encoder inputs in terms of size and channel number.

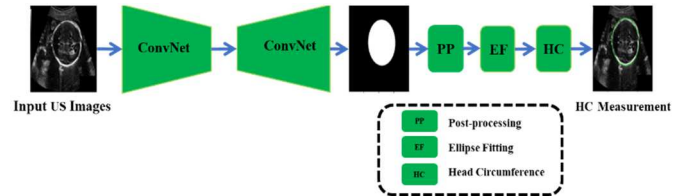


Fig. 3. Segmentation method-based conventional HC measurement method.

Finally, the segmented data is used to fit the ellipse to determine its length by obtaining its parameters (major axis, minor axis, center points). The Ramanujan estimation methodology is applied to predict the measurement of an

ellipse, represented by the symbol HC [31] wherein $h = \frac{(a-b)^2}{(a+b)^2}$, both a and b are the ellipse's long and short axes, respectively.

$$HC = \pi(a + b) \left(1 + \frac{3h}{10 + \sqrt{4-3h}}\right) \quad (6)$$

IV. RESULT ANALYSIS AND DISCUSSION

Both approaches, regression-based and segmentation-based, were trained on the Google Collaboratory platform using the backend libraries Keras and TensorFlow. We ran the programs on a GPU with 12.7 GB of RAM and a 78.2 GB disc given by Google Collaboratory. To prevent overfitting on account of the paucity of initial data for the use of HC18, we used a value of 0.5 as the percent dropout rate and kept 50% of the parameters used in recurrent CNN models. Table I shows the quantity of variables that each network may have developed.

TABLE I. AN OVERVIEW OF CNN MODELS FOR THE PROPOSED REGRESSION AND SEGNET MODELS. D1=EFFICIENTNETB2, D2=RESNET50V2, AND D3=MOBILENETV2, R=REGRESSION & M= MILLION.

Layers	Segmentation Model (Seg_Net)	Regression Models (R-D1 + R-D2 + R-D3)
Input Size	224× 24	224×224
Activation Function	ReLU & SoftMax	Linear
Learning Rate	0.0001	0.0001
Re-scaling	1/255	1/255
Batch Size	30	30
Dropout	-	0.5
Optimizer	Adam	Adam
Total Parameters	7,84,385	7.70 M + 23.51 M + 2.23 M

A. HC Measurement Results Analysis based on Deep Ensemble Regression

With five distinct pre-trained CNN backbones, we study the HC18 dataset while training and testing regression CNN architectures using three loss estimators (MAE loss, Huber loss & MSE loss). Table II presents the assessment of five distinct pre-trained CNN backbones for direct HC estimation. It is evident that the Regression ResNet50V2 model, when combined with the MSE loss, outperforms the other CNN models: The single regression network's best result MAE is 1.35 mm.

B. HC Measurement Result Analysis based on Segmentation

After the SEGNet model has been trained and tested on our dataset, we calculate the performance measures. The Dice index (DI) as well as the Hausdorff distance (HD), that reflects the greatest one point to another distance (Dist) across two stipulates, and the average symmetrical surface distance (ASSD) corresponding to the segmentation results (Seg) as well as the ground truth map (GT) are the metrics employed. The segmentation evaluation matrices' mathematical formulations are provided in (7) through (9), and the evaluation matrices' quantitative values are found as

Dice index as 98.25 ± 0.8 , Hausdorff distance as 1.33 ± 2.26 , and ASSD as 0.35 ± 0.25 , respectively.

$$Dice = \frac{2Seg.GT}{Seg+GT} \quad (7)$$

$$HD = Max \left(Max(Dist(Seg, GT)), Max(Dist(Seg, GT)) \right) \quad (8)$$

$$ASSD = Mean \left(Mean(Dist(Seg, GT)), Mean(Dist(Seg, GT)) \right) \quad (9)$$

TABLE II. THE MEAN PERFORMANCE OF FIVE CNN MODELS. MAE: MEAN ABSOLUTE ERROR AND PMAE: PERCENTAGE MAE

Models	Loss Matrices	MAE (mm)	PMAE (%)	Elapsed Time (Sec.)
Reg-InceptionV3	MAE	4.81 ± 4.90	2.71 ± 3.44	1188.24
	MSE	3.42 ± 3.58	3.07 ± 3.29	1128.27
	HL	6.54 ± 10.43	5.33 ± 9.23	1150.23
Reg-ResNet50V2	MAE	2.85 ± 2.95	2.20 ± 2.62	1314.20
	MSE	1.35 ± 1.51	1.21 ± 2.05	1298.11
	HL	2.84 ± 2.54	2.78 ± 3.03	1250.80
Reg-EfficientNetB2	MAE	3.08 ± 3.96	3.01 ± 3.40	1580.29
	MSE	2.88 ± 3.09	2.96 ± 3.32	1550.88
	HL	4.22 ± 4.34	3.76 ± 3.97	1531.45
Reg-MobileNetV2	MAE	4.19 ± 3.99	3.69 ± 4.05	822.19
	MSE	3.75 ± 3.50	3.49 ± 3.89	821.77
	HL	5.46 ± 4.87	5.10 ± 4.99	815.87
Reg-DenseNet201	MAE	13.04 ± 11.65	10.23 ± 9.43	1446.89
	MSE	6.36 ± 7.68	5.67 ± 5.29	1456.60
	HL	8.78 ± 8.60	7.37 ± 6.83	1396.90

C. Performance Comparison between the Segmentation CNN and the Regression CNN

For evaluating the results of segmentation-free versus segmentation-based methods, we collected the best three regression-based CNN model results of the MSE loss function from Table II, which were used to build our ensemble CNN regression architecture, and the SEGNet performance depicted in Table III. It is clear from this table that the SEGNet performance is better than the single CNN regression models, but the Reg-Ensemble model performance is the best of all them. The Reg-Ensemble model's and SEGNet elapsed time are respectively 1842.29 and 1877.83 seconds, for an equal number of epochs. So, the time complexity of the Reg-Ensemble model is a little bit less than the time complexity of SEGNet. Fig. 4 describes the Reg-Ensemble & SegNet based results. Furthermore, the loss curves of Fig. 5 also justify the superiority of the Reg-Ensemble Model.

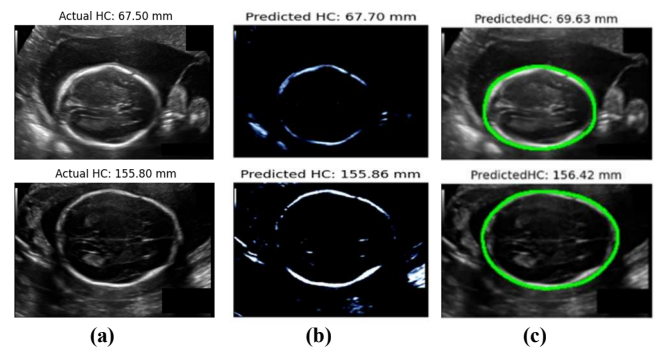


Fig. 4. (a) Original US images and their corresponding (b) Reg-Ensemble predicted HC, and (c) SegNet predicted HC

TABLE III. PERFORMANCE EVALUATION OF THE INDIVIDUAL REGRESSION MODELS, THE ENSEMBLE MODEL, AND THE SEGMENTATION MODEL OF THE HC MEASUREMENT.

Models	MAE (mm)	PMAE (%)
Reg-ResNet50V2	1.35 ± 1.51	1.21 ± 2.05
Reg-EfficientNetB2	2.88 ± 3.09	2.96 ± 3.32
Reg-MobileNetV2	3.75 ± 3.50	3.49 ± 3.89
Seg-Net	1.26 ± 1.50	1.15 ± 1.98
Reg-Ensemble Model	1.05 ± 1.22	0.96 ± 1.27

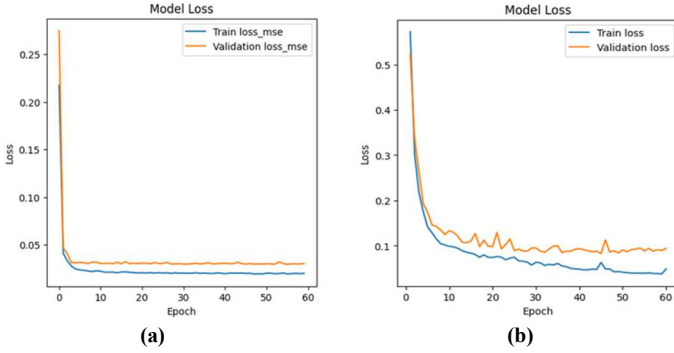


Fig. 5. Loss curves of (a) Reg-Ensemble Model and (b) SegNet Model.

We also use linear regression to examine how well the estimated HC values from the two different approaches correspond with the actual HC values. Whether the models are segmentation or ensemble regression, Fig. 6 demonstrates a remarkable linear connection between the reference and prediction values. The segmentation model tends to underestimate comparatively large HC values, as shown by the little variation in the scatter plot.

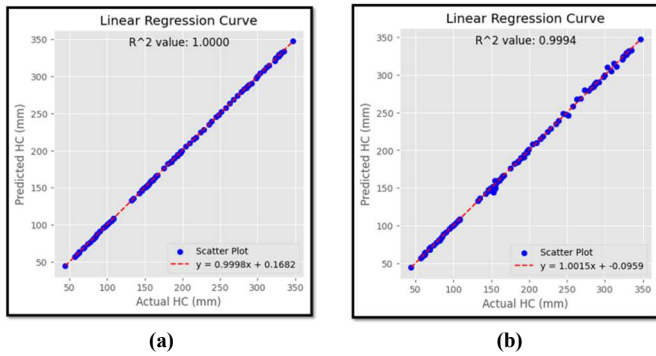


Fig. 6. Linear Regression Curve (a) Reg-Ensemble Model, and (b) SegNet Model.

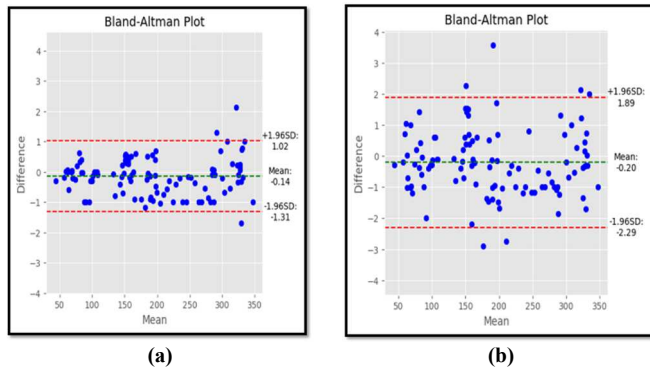


Fig. 7. Bland-Altman plot of (a) Reg-Ensemble Model and (b) SegNet Model. The ground truth and predicted HC differences (in millimeters) are displayed on the y-axis, while the mean of the two is displayed on the x-axis. The top layer and bottom layer of 95% consistency are shown by the red lines with dots, which are horizontal; the green dotted line in the middle depicts the mean difference.

Another method for evaluating the degree of agreement around two measures is the Bland-Altman plot, which makes it simple to identify bias in the data by displaying the variance between each measurement vs their mean based on the Bland-Altman plot generated on the 10% test images in Fig. 7. The ensemble regression model has 5 out of 10% test image points that go beyond the 95% agreement limit, whereas the model of segmentation has 6 out of 10% test image points, usually with higher HC values. $\{-2.29 \text{ mm}, 1.89 \text{ mm}\}$ and $\{-1.31 \text{ mm}, 1.02 \text{ mm}\}$ are the corresponding 95% agreement limits for the segmentation and ensemble regression models.

D. Comparison of the Proposed Work and the Related Works

Lastly, Table IV compares the suggested segmentation method and the suggested segmentation-free regression ensemble method with the most advanced segmentation techniques. Since the protocols of many articles differ, it is impossible to present a fully correct comparison, but we offer the results as a way to approximate the amount of the estimated inaccuracy. However, we can conclude from Table IV that the Reg-Ensemble model we suggested for this study performs noticeably.

TABLE IV. HC MEASUREMENT VIA THE HC18 OPEN ACCESSIBLE DATASET IN CONTRAST TO THE STATE OF THE ART. N/A STANDS FOR NOT APPLICABLE, AND DI FOR DICE INDEX.

Study	MAE (mm)	DI (%)
Segmentation-Based Approach		
Sobhaninia et al. [19]	2.12 ± 1.87	96.84 ± 2.89
Moccia et al. [20]	1.95 ± 1.92	97.90 ± 1.11
Fiorentino et al. [21]	1.90 ± 1.76	97.75 ± 1.32
Seg-Net used in this study following the concept of [30]	1.26 ± 1.50	98.25 ± 0.80
Segmentation-Free Approach		
J. Zhang et al. [7] Reg-Resnet50	4.78 ± 4.41	N/A
Proposed Reg-Ensemble Method	1.05 ± 1.22	N/A

V. CONCLUSION

This paper proposed and investigated a one-stage method made up of a deep combined CNN regression framework for autonomously determining the fetal size from ultrasound photographs. Comparing the ensemble deep CNN regression model and segmentation techniques was also our aim. Segmentation-based approaches yield accessible outcomes for HC prediction while their segmented result is visual; nonetheless, they frequently necessitate certain subsequent processing methods. In contrast, CNN-based regression methods are found to be more accurate in HC prediction directly without the segmentation tasks, despite their unclear interpretability. This research examines segmentation and segmentation-free techniques using advanced CNN architectures. Compared to the segmentation-based technique, the suggested regression ensemble method has a lower estimated error. Therefore, regression-based ensemble CNN techniques for direct estimation have significant potential and should be further studied in the future. But there exist some limitations, such as inability to handle outliers, limited generalization, and post-processing dependency on HC measurement. In upcoming studies, we will evaluate standard

regression CNNs on additional clinical data sets for determining numerous biomarkers. In addition, we intend to explore segmentation-free methods for simultaneous learning, which integrates both the segmentation section and regression section, and other, more recent CNN frameworks with better feature illustration capabilities.

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